

A 3×3 Illustration of Midpoint IDW and Link-Distance Weighting

1 IDW Underestimate of Rain Rate

We use a 3×3 field whose cell side length is h and whose rain rates, listed from top to bottom, are

$$R^{\text{GT}} = \begin{bmatrix} 4 & 4 & 4 \\ 5 & 5 & 5 \\ 4 & 4 & 4 \end{bmatrix} \text{ mm/h.} \quad (1)$$

To obtain a valid example for the normalized IDW definition, place one horizontal link through the center of each row. The link-level rain rates are then 4, 5, and 4 mm/h, respectively. We denote the middle link by ℓ and examine the three pixels that it crosses. All three links have their midpoint in the middle column, and the IDW search radius is chosen larger than $\sqrt{5}h$, so every link contributes at every non-midpoint cell.

For a pixel center c_p that does not contain a link midpoint, midpoint IDW with exponent two gives

$$\hat{R}_p^{\text{IDW}} = \frac{\sum_{j=1}^3 \|c_p - m_j\|_2^{-2} R_j}{\sum_{j=1}^3 \|c_p - m_j\|_2^{-2}}, \quad (2)$$

where m_j and R_j are the midpoint and rain rate of link j . At the center of the left or right pixel crossed by ℓ , the distances to the middle, upper, and lower link midpoints are h , $\sqrt{2}h$, and $\sqrt{2}h$. Therefore,

$$\hat{R}_1^{\text{IDW}} = \hat{R}_3^{\text{IDW}} = \frac{5h^{-2} + 4(\sqrt{2}h)^{-2} + 4(\sqrt{2}h)^{-2}}{h^{-2} + 2(\sqrt{2}h)^{-2}} = 4.5 \text{ mm/h.} \quad (3)$$

The middle pixel contains the midpoint of ℓ , so the discretized midpoint rule sets its value directly to $R_\ell = 5$ mm/h. Repeating the same calculation for the remaining cells gives

$$\hat{R}^{\text{IDW}} = \begin{bmatrix} 73/17 & 4 & 73/17 \\ 9/2 & 5 & 9/2 \\ 73/17 & 4 & 73/17 \end{bmatrix} \approx \begin{bmatrix} 4.29 & 4 & 4.29 \\ 4.5 & 5 & 4.5 \\ 4.29 & 4 & 4.29 \end{bmatrix} \text{ mm/h.} \quad (4)$$

Thus, along ℓ , midpoint IDW underestimates the true rain rate in the two off-center pixels, although it is exact in the midpoint pixel.

ILDW instead uses the distance from a pixel center to the full link segment and applies the crossing-pixel rule. If $C(p)$ denotes the set of links that cross pixel p , then

$$\hat{R}_p^{\text{ILDW}} = \frac{1}{|C(p)|} \sum_{j \in C(p)} R_j, \quad C(p) \neq \emptyset. \quad (5)$$

Every cell in a given row is crossed by that row's link. Consequently ILDW recovers the complete field in this example,

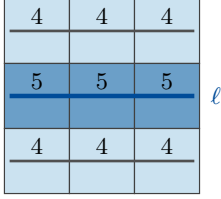
$$\hat{R}^{\text{ILDW}} = R^{\text{GT}}, \quad (6)$$

and its profile along ℓ is uniformly 5 mm/h.

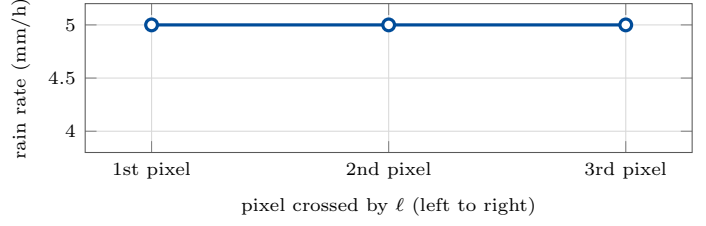
Why the literal single-link version does not work. If ℓ is the only link and $R_\ell = 5$ mm/h, then for every in-radius non-midpoint pixel the normalized IDW weight cancels:

$$\hat{R}_p^{\text{IDW}} = \frac{\|c_p - m_\ell\|_2^{-2} R_\ell}{\|c_p - m_\ell\|_2^{-2}} = R_\ell = 5 \text{ mm/h.} \quad (7)$$

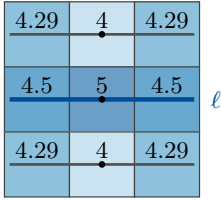
The midpoint rule gives the same value in the middle pixel. Hence a setup containing only one link cannot demonstrate IDW underestimation under the normalized IDW definition; the two auxiliary links above are the minimal addition that produces the intended comparison. Alternatively, one would have to replace normalized IDW by an unnormalized distance-decay kernel, which would be a different estimator.



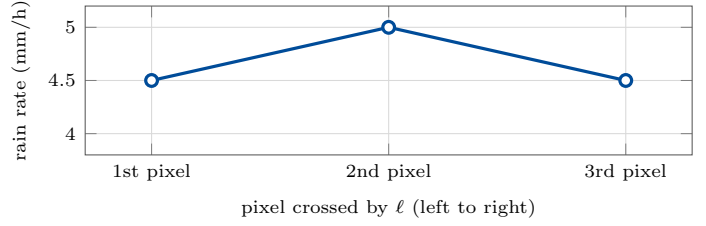
(a) Ground-truth field and link geometry.



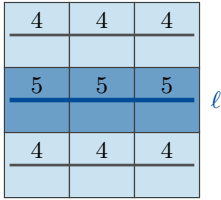
(b) Ground-truth profile along ℓ .



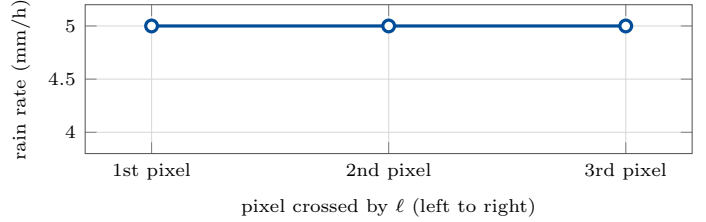
(c) Midpoint-IDW reconstruction. Dots mark the three virtual gauges.



(d) Midpoint-IDW profile along ℓ .



(e) ILDW reconstruction.



(f) ILDW profile along ℓ .

Figure 1: A concrete 3×3 example of the geometric difference between midpoint IDW and ILDW. Midpoint IDW mixes the 5 mm/h observation on the middle link with the two nearby 4 mm/h observations away from the middle pixel, producing (4.5, 5, 4.5) mm/h along ℓ . Since ILDW associates the link observation with every pixel crossed by its segment, it returns (5, 5, 5) mm/h along ℓ and exactly reconstructs this row-wise-constant field.